Epsilon Carve v0.5.2 — User Manual

Nov 8, 2026 · @Edison Muzio

Welcome

Epsilon Carve makes room for one sound inside another, and shows you the result as a face coming out from behind a mask. Put it on the beat, feed it the vocal, and it dips the beat only at the frequencies where the beat is covering the vocal, only while the vocal is singing. Nothing is delayed: Carve has zero latency, so you can record and monitor through it.



The problem it solves: masking

When two sounds share the same frequencies, the louder one hides the quieter one. Engineers call this **masking**. A vocal at 1 kHz disappears under a synth pad or a guitar at 1 kHz, even when the vocal fader is up. Turning the vocal up makes the mix louder, not clearer. The fix is to carve a little out of the beat, exactly where and when the vocal needs it. Doing that by hand means dozens of automated EQ moves. Carve re-decides every cut 3,000 times a second across 32 frequency bands.

What makes Carve different

Feature	What it means for you
Collision mode	Carve listens to both tracks. It cuts only where the beat is actually loud enough to bury the vocal, and only as deep as needed. The beat stays fuller.
Unmask face	The vocal is a face; the beat is a mask over it. As you raise Amount, the mask lifts off piece by piece. You see clarity, not just EQ curves.
Width Guard	Carves the centre of the beat, where the vocal lives, and leaves the stereo sides untouched. Wide beats stay wide.
Zero latency	Safe on a live vocal chain, while tracking, and on a laptop.

Quick start: 60 seconds

1. Put **Epsilon Carve** on the beat (or the instrument bus) you want to make room in.
2. Route the **vocal** into Carve's sidechain input (Setup, next section, shows how in FL Studio, Logic and Ableton).
3. Play the song. The badge on the display turns cyan and reads **SIDECHAIN LIVE**, and the face lights up.
4. Pick the **Vocal Space** preset.
5. Turn **Amount** up slowly from 0% while the vocal plays. Watch the mask lift off the face and the **Clarity** number rise. Stop when the vocal sits clearly in front; most mixes land between 30% and 60%.
6. Toggle bypass at matching loudness to confirm it sounds better, not just different.

Setup

Carve goes on the sound that should make room; the sound that needs room goes into its sidechain. Get that one routing right and everything else is turning knobs.

Install

1. Build it from the Projucer project (double-click `Epsilon-Carve.jucer`, Save and Open in IDE, press ⌘B on **Epsilon Carve - VST3** and **AU**). Xcode copies the plug-ins into `~/Library/Audio/Plug-Ins/VST3` and `~/Library/Audio/Plug-Ins/Components`.
2. Rescan plug-ins in your DAW. Carve appears under **Epsilon Sounds**.

Routing the sidechain

DAW	Put Carve on	Send the vocal by	Then
FL Studio	The beat's Mixer insert	Select the vocal insert, right-click the routing arrow under the beat insert, choose **Sidechain to this track**	In Carve's wrapper: gear menu → **Processing → Connections**, assign the vocal to **Sidechain**
Logic Pro	The beat track or bus (Audio Units)	Carve's header: **Side Chain** menu → choose the vocal track	Nothing else
Ableton Live	The beat track (VST3 or AU)	Open Carve's device, expand **Sidechain**, enable it, pick the vocal track, **Post FX**	Keep the vocal's own output going to the Master
Any other DAW	The track to carve	Its sidechain or auxiliary input feature	Look for the badge to read SIDECHAIN LIVE

Keep the vocal's normal route to the master. The sidechain is a copy for Carve to listen to; it is never heard through Carve.

Reading the status badge

The badge at the top right of the display tells you whether the routing works.

Badge	Meaning	What to do
SIDECHAIN LIVE (cyan)	The vocal is arriving.	Nothing; you're set.

Badge	Meaning	What to do
SIDECHAIN QUIET	The input is connected but silent.	Play a section with vocal, or check the send level.
NO SIDECHAIN ROUTED	The host has not connected anything to the sidechain input, so Carve listens to the beat itself.	Redo the routing step for your DAW.
INTERNAL SIDECHAIN	You switched Carve to Internal on purpose.	Switch to External if you meant to use the vocal.
WAITING FOR AUDIO	Transport stopped, or no audio reaching Carve.	Press play.

Tour of the panel

The panel reads top to bottom: choose a preset and mode, watch the face and the display, then shape the cut with the knobs. The numbers below match the numbered markers on the annotated screenshot at the end of this section.

#	Part	What it is for
1	**Preset bar**	Ten starting points. The arrows step through them; the menu jumps straight to one.
2	**Collision / Classic**	How Carve decides where to cut. Collision is the default.
3	**Unmask face**	The sidechain (face) under the track (mask); the face is a frequency scale, 20 kHz at the top. Shards lift as each slice is cleared. Clarity % and the Making room for picker.
4	**Collision display**	Beat spectrum (amber), vocal spectrum (cyan), collision zones (coral) and the exact cut curve (white, lower graph).
5	**Status badge**	Whether the sidechain routing works.
6	**Range handles**	Drag left and right to set Low Cut and High Cut directly on the graph.

#	Part	What it is for
7	**Sidechain: External / Internal**	Use the routed vocal, or let the beat carve itself.
8	**Show / Hide Advanced**	Switches between Simple view (3 knobs) and Advanced view (7 knobs + Width Guard + Delta).
9	**Width Guard**	Carve the centre only; keep the stereo sides. *(Advanced)*
10	**Delta**	Hear only what Carve removes. *(Advanced, and always visible while on)*
11	**Amount**	The main control: how hard Carve cuts.
12	**Low Cut / High Cut**	The frequency range Carve may touch.
13	**Sensitivity, Attack, Release, Mix**	Fine control of how Carve reacts. *(Advanced)*

14 · A / B: two complete settings to compare, saved with the session (see A/B and edited presets under Every control explained).

Simple view and Advanced view

Carve opens in **Simple view**: Low Cut, Amount and High Cut, the mode switch and presets. That is everything most sessions need, and it keeps the face and display large and calm. Click **Show Advanced** for the other four knobs, Width Guard and Delta. Carve saves the view with your session, so a project you finished in Advanced view reopens in Advanced view.



The window scales from 85% to 200%: drag its corner. The whole panel scales as one piece, so the layout and text never rearrange.

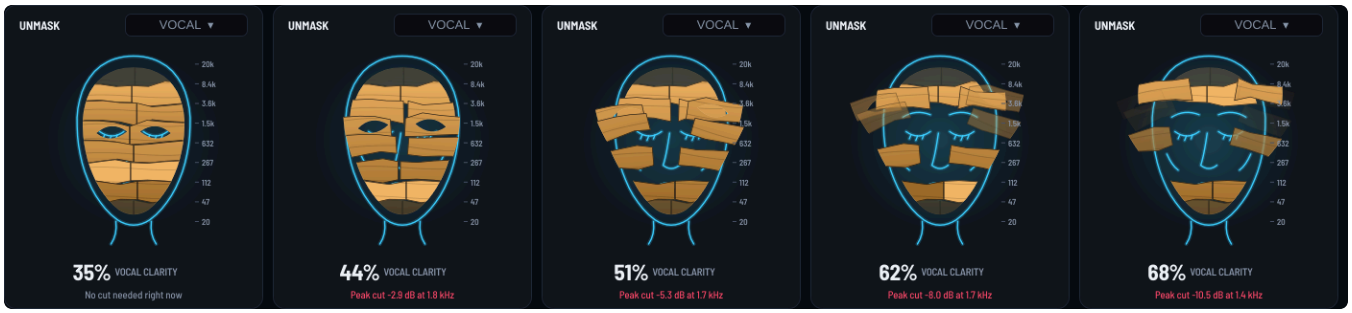


The Unmask face

The face is the sound that needs room, the mask is the track covering it, and the face is a frequency scale: each shard is one slice of the spectrum and lifts by exactly how much of that slice's masking Carve has removed. It turns frequency masking into something you can read in half a second from across the room.

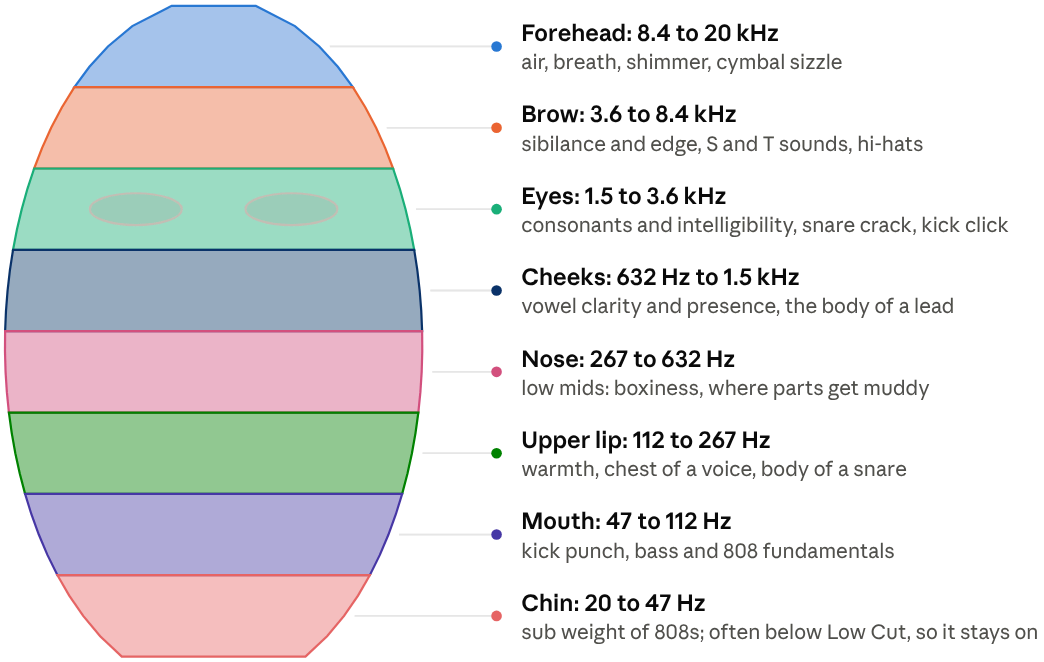
- **The face** is the sidechain, drawn in the sidechain's cyan. Name it with the picker at the top right of the panel: **Vocal, Kick, Snare, 808 / Bass, Lead, Voice-over** or **Other**. The picker changes the labels ("KICK CLARITY") and suggests a matching preset; it never changes the sound.
- **The mask** is the track, in the track's amber.
- **Top to bottom is 20 kHz down to 20 Hz** on a log scale: eight equal slices of 1.25 octaves each, with the slice edges printed beside the face (diagram below).
- **Each shard follows only its own slice.** Half of that slice's masking removed = the shard half lifted. Nothing borrowed from neighbouring slices.
- **A faint, still shard** means the sidechain has almost nothing in that slice, so there is nothing to unmask there.

- **Clarity %** is how much of the sidechain is clear of the track right now, weighted by where the sidechain is loudest. **Peak cut** shows the deepest cut and where it is.



The face is a frequency scale: each shard is one slice

Eight equal 1.25-octave slices, 20 kHz at the top to 20 Hz at the chin. Each lifts by the share of its masking removed.



the mask's eight shards and the frequency range each one covers

How to read it

What you see	What it means	What to try
Whole mask, Amount at 0%	Carve is not cutting; the mask shows the masking that's there.	Raise Amount.

What you see	What it means	What to try
Shards lifting from the eyes and cheeks	The 632 Hz–3.6 kHz slices are being cleared: for a vocal, the words get easier to follow.	This is usually the sweet spot. Stop when the words are clear.
Forehead or chin shards never move	Those ranges are outside your Low Cut / High Cut, so Carve isn't allowed to cut there.	Widen the range if you want them cleared, or leave them: often that's intentional.
A shard lifts, then settles back	The beat got busier in that range (a chorus, a new synth).	Normal. Collision follows the music.
Face dim, "Route a sidechain"	No vocal is reaching the sidechain.	See Setup.
"DELTA ON" under the face	You are monitoring only what is removed.	Turn Delta off before bouncing.

How it is calculated

In each of Carve's 32 bands, Carve compares the track's level, after the cut, with the sidechain's level. The sidechain counts as **clear** when the track is 9 dB or more below it, the same clearance Collision mode aims for, and **buried** when the track is 3 dB or more above it; in between it is partly masked. Bands are weighted by how loud the sidechain is there, ignoring anything more than 36 dB below its loudest band. A band whose frequency span crosses a slice edge is shared between the two slices in proportion, so no band is rounded into the wrong shard.

Clarity is that weighted average, as a percentage. **Each shard** compares its slice's masking now with what it would be with no cut at all, and lifts by the share removed, linearly. Shards lift quickly and settle back slowly, and the face holds its last state while the sidechain is silent.

The Collision display

The upper graph shows both tracks and where they collide; the lower graph shows exactly what Carve is cutting. The face tells you how much; the display tells you where.

Element	Colour	Meaning
Track spectrum	Amber line and fill	The beat, before Carve touches it.
Sidechain spectrum	Cyan line with a soft glow	The vocal.
Collision zones	Coral columns rising from the floor	Frequencies Carve is cutting right now. Brighter = deeper; a fine diagonal hatch marks cuts of 1 dB or more, so zones read without relying on colour.
Cut graph	White line over a coral body	The real cut in dB, 0 to −18, calculated from the live filters (all 32 bands and Mix included).
Dashed cyan line	Lower graph, only with Width Guard on	The cut on the stereo sides. Flat at 0 dB = width untouched.
Shaded areas	Dark, outside the handles	Never cut: outside Low Cut / High Cut.
White handles	Top of the graph	Drag to set Low Cut and High Cut.

Both spectra come from a 4096-point analyser smoothed to 1/6 octave, so you see individual notes and harmonics, not just a rough shape. They rise instantly and fall at about 25 dB per second, so peaks stay readable.

Hover readout

Hover anywhere over the graphs and the line under them reads, for that exact frequency: **track level, sidechain level and the cut**. Use it to answer "why is the vocal fighting at 2.4 kHz?" in one move: if the track is above the sidechain there, that is your collision.

Dragging the range

The white handles are Low Cut and High Cut. Grab one within a few pixels and drag; the cursor changes to a left-right arrow when you are on it. Moves are recorded as normal automation gestures, so you can automate the range by dragging while your DAW is in write mode.

Every control explained

Amount does most of the work; the range knobs say where; the Advanced knobs say how fast and how sensitive. Double-click any knob to reset it. Drag up/down or left/right, or use the mouse wheel. Click the value under a knob to type a number (frequencies accept `k`, so `2.5k` = 2.5 kHz).

Control	Range · default	What it does	Why you'd move it	Start at
Amount	0–100% · 25%	How hard Carve cuts. At 100% a band can ask for up to 18 dB.	More clarity (raise) or more natural beat (lower).	30–60%
Low Cut	20 Hz–20 kHz · 20 Hz	Lowest frequency Carve may cut.	Protect kick and 808 weight while clearing above.	100–150 Hz for vocals
High Cut	20 Hz–20 kHz · 20 kHz	Highest frequency Carve may cut.	Keep hi-hats, cymbals and air untouched.	6–10 kHz for vocals
Collision / Classic	switch · Collision	How Carve decides each cut (next section).	Classic for punchy, every-hit ducking.	Collision
Sidechain	External / Internal · External	What drives the cuts: the routed vocal, or the track itself.	Internal turns Carve into a self-taming dynamic EQ.	External
Sensitivity	−48 to +24 dB · 0 dB	How strongly the sidechain counts. In Collision it also asks for a little more clearance.	Raise for quiet or distant vocals; lower if breaths and bleed trigger cuts.	0 dB
Attack	1–150 ms · 15 ms	How fast a cut arrives.	Short (1–5 ms) for drums; 10–20 ms for vocals keeps the beat's transients.	10 ms

Control	Range · default	What it does	Why you'd move it	Start at
Release	10–1000 ms · 180 ms	How fast the beat recovers.	Short follows fast syllables; long sounds smoother and less noticeable.	120–250 ms
Mix	0–100% · 100%	Blend of carved and original.	Below 100% gives a gentler, parallel version of the cut.	100%
Width Guard	on / off · off	Cut the centre (Mid) only, keep the sides.	Wide pads, stereo reverbs, wide synths.	On for wide beats
Delta	on / off · off	Hear only what Carve removes.	Checking that the cut is in the right place.	Off

Gotchas

- **Low Cut above High Cut** closes the range: nothing is cut and the display says so.
- **Attack and Release use slow-in, fast-at-the-bottom knobs.** Most of the knob's travel covers the useful short times; the long times sit at the far end.
- **Amount 0% is a true bypass of the processing:** output is bit-for-bit identical to the input. So is Collision mode with a silent sidechain.
- **Delta exports only the cut** if you bounce with it on. The display and face warn you while it is on.

A/B and edited presets

- **A / B** sits next to the Sidechain button. Click **B** to try a different setting without losing **A**, and click **A** to go back. The arrow (A→B or B→A) copies the active slot to the other. Both slots are saved with the session.
- An **asterisk** after the preset name ("Vocal Space *") means a control has moved since the preset was loaded. Pick the preset again to return to it. Delta never counts as an edit.

The modes: Collision, Classic, Width Guard, Delta, Internal

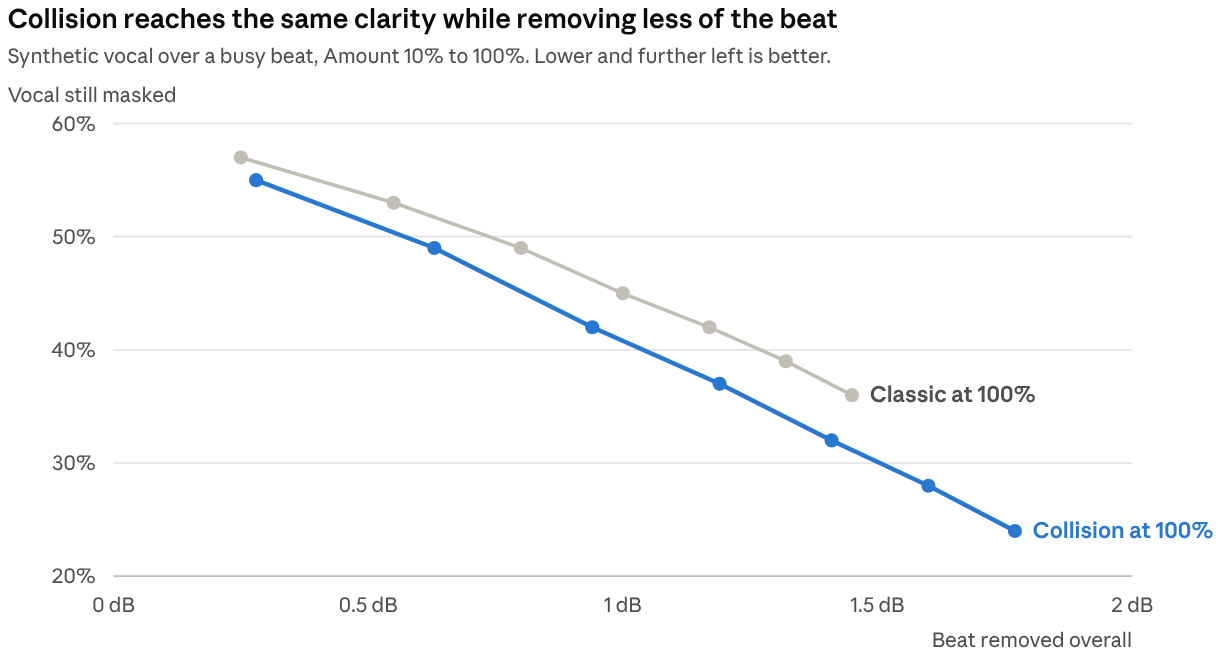
Use Collision for vocals and leads, Classic for drums, Width Guard for wide beats, and Delta to check your work.

Collision (default)

Collision measures **both** tracks in every band. It asks: is the beat loud enough here to bury the vocal? If not, the band is left alone. If yes, it cuts just enough for the vocal to sit about **9 dB clear** of the beat there. So the beat keeps everything the vocal doesn't need, and in quieter sections Carve backs off by itself.

Classic

Classic cuts in proportion to how loud the vocal (sidechain) is in each band, whatever the beat is doing. It is predictable and punchy: every kick hit carves the same notch out of the 808. Use it for **kick vs 808**, **snare pockets**, and anywhere you want the classic "duck on every hit" feel.



Carve v0.5.0 engine, offline render of a synthetic vocal over a beat, 48 kHz; not a listening test

In the test above, Collision at about 55% gave the same clarity as Classic at 100% while removing 18% less of the beat, and it kept going to clarity Classic cannot reach. When the beat dropped 15 dB (a breakdown), Collision removed 69% less than Classic, because nothing was in the way.

Width Guard

Carve splits the beat into **Mid** (what's in the centre, where the lead vocal sits) and **Side** (what's spread left and right: pads, room, stereo synths). With Width Guard on, only the Mid is carved. The beat keeps its full width while the vocal gets room in the middle. Switching it glides over 30 ms, so it never clicks. The dashed cyan line on the cut graph shows the Side cut: flat means the width is untouched.

Delta

Delta plays **only what Carve is removing**. It is the fastest way to hear whether the cut is in the right place: you should hear a thin, ghostly version of the beat that follows the vocal's rhythm. If you hear the kick or bass thumping, raise Low Cut; if you hear hi-hats, lower High Cut. While Delta is on the display border pulses coral, a banner says so, the face reads DELTA ON, and the Delta button stays visible in Simple view.



Internal sidechain

With **Sidechain: Internal**, the track drives its own cuts: Carve becomes a 32-band dynamic EQ that dips whatever is loudest in the track at that moment. It is a quick fix for a harsh or boomy bus with no routing at all. Collision needs two different signals, so in

Internal mode Carve uses Classic behaviour automatically.

Presets

Every preset is a starting point: pick the closest one, then adjust Amount first and the range second. Presets also appear in your DAW's program list, so you can switch them from the host or automate them.

Preset	Use it for	Mode	Amount	Range	Attack / Release	Width Guard	First tweak
Init	A neutral start	Collision	25%	20 Hz–20 kHz	15 / 180 ms	off	Set the range
Vocal Space	Lead vocal over a full beat	Collision	40%	120 Hz–8 kHz	10 / 160 ms	on	Amount until the words are clear
Rap Vocal Pocket	Fast, dense rap delivery	Collision, Sens +3 dB	50%	150 Hz–6 kHz	5 / 120 ms	on	Shorten Release if syllables smear
Lead Over Beat	Synth or guitar lead	Collision	35%	250 Hz–7 kHz	12 / 200 ms	on	Move Low Cut to the lead's lowest note
Kick Through 808	Kick on the sidechain, Carve on the 808	Classic	55%	30–250 Hz	2 / 90 ms	off	Release to the length of the kick
Snare Pocket	Snare on the sidechain, Carve on pads or guitars	Classic	35%	150 Hz–5 kHz	1 / 80 ms	off	Amount for the crack you want
Clean Low Mids	Mud between two parts	Collision	45%	200–700 Hz	15 / 220 ms	off	Slide both handles to the muddy spot
Voice-Over Bed	Podcast, ad or film voice over music	Collision, Sens +6 dB	70%	100 Hz–10 kHz	5 / 300 ms	off	Longer Release for a smoother bed
Air For The Hook	Presence and air for a hook vocal	Collision	35%	3–16 kHz	8 / 150 ms	on	Watch the brow and forehead shards
Gentle Glue	Light, always-on separation	Collision	15%	20 Hz–20 kHz	20 / 250 ms	off	Leave it subtle

Changing a knob after loading a preset keeps the preset's name showing; your changes are saved with the session either way.

Recipes

Six step-by-step workflows for the situations Carve was built for.

1. Vocal sits behind the beat

1. Carve on the beat bus, vocal into the sidechain, preset **Vocal Space**.
2. Raise **Amount** while the vocal plays until the eye and cheek shards (1–4 kHz) lift. That is where the words live.
3. If the low end of the beat feels thinner, raise **Low Cut** until it comes back (often 150–200 Hz).
4. Turn on **Delta** for five seconds: you should hear a thin, rhythmic ghost of the beat. Turn it off.
5. A/B with bypass at the same loudness.

2. Kick and 808 fighting

1. Carve on the **808**, the **kick** into the sidechain, preset **Kick Through 808** (Classic).
2. Set **Release** so the 808 comes back just as the kick's tail ends (60–120 ms).
3. Raise **Amount** until the kick's click and thump punch through without the 808 pumping audibly.
4. For sub-heavy 808s, lower **High Cut** to 120–150 Hz so only the sub is ducked.

3. Rapper with fast delivery

1. Preset **Rap Vocal Pocket**.
2. Shorten **Release** to 80–120 ms so the beat recovers between syllables instead of staying ducked.
3. If quiet ad-libs don't trigger cuts, raise **Sensitivity** 3–6 dB.

4. Wide beat, narrow vocal

1. Any vocal preset, then **Show Advanced** and turn on **Width Guard**.
2. Raise Amount. The beat's stereo pads and reverbs stay wide; only the centre makes room.

3. Check the dashed cyan line: flat at 0 dB means the sides are untouched.

5. Voice-over on a music bed

1. Carve on the music, voice into the sidechain, preset **Voice-Over Bed**.
2. Use Amount 60–80%. Longer Release (300–500 ms) makes the music swell back gently between sentences.
3. Because Collision only cuts where the music competes with the voice, the bed keeps its low end and sparkle.

6. A harsh or boomy bus, no routing

1. Switch **Sidechain** to **Internal**.
2. Set the range around the problem (for harshness 2–6 kHz, for boom 100–300 Hz) by dragging the handles onto it.
3. Raise Amount slowly. Carve now dips that range only when it gets loud: a dynamic EQ in three moves.

Creator's notes

These are the behaviours that come from how Carve is built, and how to use them on purpose.

Hidden behaviours worth knowing

What Carve does under the surface	How to use it
In Collision mode, **Sensitivity also sets the clearance**. Each +1 dB asks for 1 dB more space between vocal and beat (default 9 dB).	Treat Sensitivity as a "how much space" knob in Collision: +3 to +6 dB for a vocal that must cut through a dense chorus.
Carve **ignores any part of the vocal more than 36 dB below its loudest band.**	Breaths, room tone and reverb tails don't trigger cuts, so you rarely need to lower Sensitivity for them.
Cut depth is **smoothed over 4 ms** on top of Attack.	Attack below ~4 ms gives no faster cut, just a faster detector. For drums, 1–2 ms is already the fastest it gets.

What Carve does under the surface	How to use it
Carve has **no look-ahead** (that is why it has zero latency).	On very sharp transients, the cut arrives a few ms after the hit. If a kick's click still pokes through Classic mode, that's why: shorten Attack and accept it, or nudge the kick track a few ms early.
The range handles **fade bands in and out continuously**, even between two band centres.	You can park a handle exactly on a problem frequency; you are not snapped to the 32 bands.
Delta + output = input, to within floating-point rounding.	A null test is built in: Delta is the precise removed signal, useful for checking with a phase meter or for creative routing (below).
At **Amount 0%**, and in Collision with a silent sidechain, the output is **bit-identical** to the input.	Leave Carve on a bus all the time; it adds no colour when it isn't working.
A stereo sidechain counts the **louder of left and right** in each band.	A hard-panned double still triggers cuts, so you don't need to sum it to mono first.
The **analysers and face only run while the window is open.**	In a heavy session, close Carve's window: it saves roughly 1% of a core per instance.
The face's shards **leave fast and return slowly**, and the face holds its last state when the vocal stops.	It's built to show the result, not flicker. Judge it during sung phrases.
Dragging a range handle **writes automation** like turning a knob.	Put your DAW in write mode and draw range moves straight on the graph: wider in the verse, narrower in the hook.
Presets are also **host programs**.	Automate preset changes per section in DAWs that support program changes.

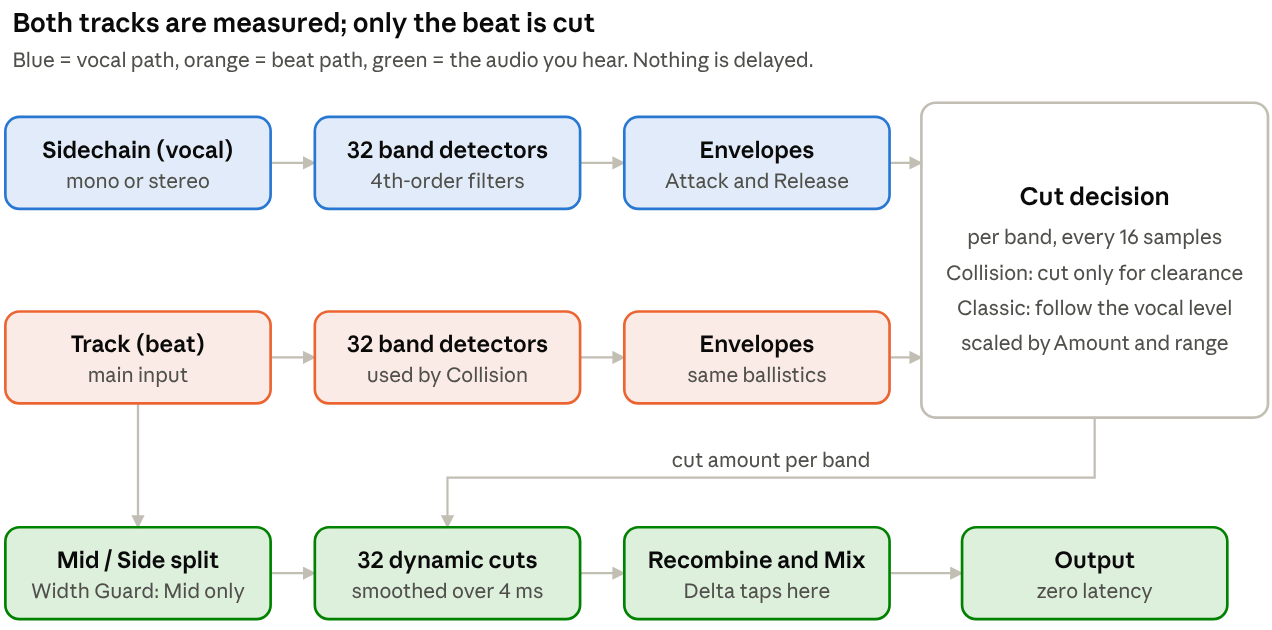
Pro tips

- **Set Amount with the face, confirm with your ears.** The face shows where masking is, but the right amount is the one where the mix sounds best, often before every shard is gone.

- **Use the range more than Amount.** A narrow range (for example 1–5 kHz) at a higher Amount is usually more transparent than a wide range at a low one.
- **Two Carves beat one big one.** Put one in Classic on the 808 keyed by the kick, and one in Collision on the music bus keyed by the vocal.
- **Creative Delta:** duplicate the beat, put Carve on the copy with Delta on, and send that copy to a long reverb. The reverb now blooms only with the parts the vocal pushes aside, a halo that follows the singer.
- **Key from a bus, not one track.** Send all lead and double vocals to a vocal bus and key Carve from the bus, so every part gets room.
- **Width Guard + Classic on kick vs bass:** the sub is mono anyway, so Width Guard keeps any stereo bass texture intact while the centre ducks.
- **Gain match before judging.** Carving lowers the beat slightly; a fair A/B needs the bypassed version turned down to match.

Under the hood

Carve measures both tracks in 32 bands, decides a cut for each band, and passes only the beat through 32 moving cuts, with no delay anywhere.



Epsilon Carve v0.5.0 signal flow

Stage	Detail
Bands	32, spaced evenly on a log scale from 20 Hz to 20 kHz (about three per octave). Bands above 47.5% of the sample rate are switched off rather than squeezed together.
Detectors	Two cascaded band-pass filters per band (4th order, Q 2.5): a loud note reads about 63 dB lower four octaves away, so it doesn't trigger distant bands. Each band's attack and release last at least two cycles of its frequency, and its cut moves are smoothed over at least one cycle: on a 40 Hz tone this cut the distortion from −42 dB to −75 dB.
Decision	Every 16 samples (0.33 ms at 48 kHz). Collision cuts the amount needed for 9 dB clearance (+ Sensitivity), capped by a ceiling that follows the vocal 3x more readily than Classic; bands more than 36 dB below the vocal's loudest are ignored.
Cuts	32 cascaded state-variable filters (trapezoidal, after Andrew Simper), Q 2, with fixed frequencies so they cannot become unstable while moving. Bandwidth is held constant in octaves up to the top of the range, so a cut at 12 kHz is as deep as one at 1 kHz.
Evenness	Cut depth for a single note stays within about 4 dB from 55 Hz to 18 kHz at 44.1, 48 and 96 kHz.
Stereo	Mid/Side processing. Width Guard fades the Side cut out over 30 ms.
Output	Recombined to left/right, Mix blended, zero latency. At Amount 0 the output is bit-identical to the input.
CPU	About 2–4% of one core at 48 kHz stereo on the build machine; about 1% more while the window is open.

The masking-aware idea follows published work on masking-reduction EQ: Hafezi and Reiss, *Autonomous multitrack equalization based on masking reduction*, Journal of the Audio Engineering Society, 2015.

Troubleshooting and FAQ

Most problems are routing; the badge and the face tell you which.

Symptom	Likely cause	Fix
Nothing happens when I raise Amount	No sidechain arriving, or Low Cut above High Cut	Check the badge reads SIDECHAIN LIVE; check the range
Badge says NO SIDECHAIN ROUTED	The host isn't feeding Carve's sidechain input	FL Studio: Processing → Connections. Logic: Side Chain menu. Ableton: Sidechain section
The beat pumps	Release too short or Amount too high	Lengthen Release to 200–300 ms; lower Amount
The low end got thin	Low Cut too low	Raise Low Cut to 120–200 Hz
Cymbals sound dull when the vocal sings	High Cut too high	Lower High Cut to 6–8 kHz
The beat sounds weird and thin everywhere	Delta is on	Turn Delta off (the display pulses coral while it's on)
Mask never fully lifts	Ranges outside Low/High Cut, or a very loud beat	Expected. Widen the range or raise Amount; judge by ear
Face and display are frozen	Transport stopped, or the vocal is silent	Press play during a sung phrase
Plug-in lags on an older Mac	A Debug build without optimisation	v0.5.0's project builds Debug at -O3; rebuild from the included Xcode project

FAQ

Does Carve add latency? No. It reports zero latency and has no look-ahead.

Does it colour the sound when it isn't cutting? No. At Amount 0, or with a silent sidechain in Collision mode, the output is bit-identical to the input.

Mono or stereo? Both, for the track and the sidechain.

Will my v0.3/v0.4 sessions still load? Yes. Parameter IDs have not changed since v0.3.0; new settings start at their defaults.

Which sample rates? Tested at 22.05, 44.1, 48, 96 and 192 kHz.

Specifications

Formats	VST3, AU (macOS 12 or later, Intel and Apple Silicon)
Latency	0 samples
Bands	32, 20 Hz–20 kHz
Maximum requested cut	18 dB per band at 100%
Window	960 × 620, scalable 75–200%
Testing	pluginval strictness 8 at 44.1/48/96 kHz; 18 engine tests (Linux build). Mac and FL Studio listening pass pending.

Credits

Epsilon Carve is designed by Edison for **Epsilon Sounds**. The face artwork and interface are original. Font: Barlow Semi Condensed, © The Barlow Project Authors, SIL Open Font License 1.1.